Week-5

2 What is the output provided by a convolution layer with the following properties:

- Stride of 1.
- filter is given by:

$$K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution:

Input image I:

$$I = \begin{bmatrix} 20 & 35 & 35 & 35 & 35 & 20 \\ 29 & 46 & 44 & 42 & 42 & 27 \\ 16 & 25 & 21 & 19 & 19 & 12 \\ 66 & 120 & 116 & 154 & 114 & 82 \\ 74 & 216 & 174 & 252 & 172 & 112 \\ 70 & 210 & 170 & 250 & 170 & 110 \end{bmatrix}$$

Now,

output size = 4X4

Row 1

$$C(1,1) = \begin{bmatrix} 20 & 35 & 35 \\ 29 & 46 & 44 \\ 16 & 25 & 21 \end{bmatrix}$$

Sum (excluding 46): $20+35+35+29+44+16+25+21 = 255$

- $$C(1,2) = \begin{bmatrix} 35 & 35 & 35 \\ 44 & 42 & 42 \\ 21 & 19 & 19 \end{bmatrix}$$

Exclude 42: $35+35+35+44+42+21+19+19 = 250$

- $C(1,4) = \begin{bmatrix} 35 & 35 & 20 \\ 42 & 42 & 27 \\ 19 & 19 & 12 \end{bmatrix}$

Exclude 42: $35+35+20+42+27+19+19+12 = 209$

Row2

$C(2,1) = \begin{bmatrix} 29 & 46 & 44 \\ 16 & 25 & 21 \\ 86 & 120 & 116 \end{bmatrix}$

Exclude(25) $= 29+46+44+16+21+86+120+116$
$= 458$

- $C(2,2) = \begin{bmatrix} 46 & 44 & 42 \\ 25 & 21 & 19 \\ 120 & 116 & 154 \end{bmatrix}$

Exclude(21) $= 46+44+42+25+19+120+116+154$
$= 566$

$C(2,3) = \begin{bmatrix} 44 & 42 & 42 \\ 21 & 19 & 19 \\ 116 & 154 & 114 \end{bmatrix}$

$= 44+42+42+21+19+116+154+114$
$= 552$

$C(2,4) = \begin{bmatrix} 42 & 42 & 27 \\ 19 & 19 & 12 \\ 154 & 114 & 82 \end{bmatrix}$

Exclude 19: $42+42+27+19+12+154+114+82$
$= 472.$

Similarly:

Row 3

$C(3,1)$

Exclude 120: $18+25+21+88+116+74+216+174$
$= 708$

$C(3,2)$

Exclude 116: $25+21+19+120+154+216+174+252$
$= 981$

$C(3,3)$

Exclude 154: $21+19+19+116+114+174+252+172$
$= 887$

$C(3,4)$

Exclude 114: $19+19+12+154+82+252+172+112$
$= 802$

Row 4

$C(4,1)$

Exclude 216: $66+120+116+74+174+70+210+174$
$= 1000$

C(4, 2)

Exclude 174 : 120 + 116 + 154 + 216 + 252 + 210 + 170 + 250

= 1488

C(4, 3)

Exclude 252 : 116 + 154 + 114 + 174 + 172 + 170 + 250 + 170

= 1320

C(4, 4) :

Exclude 172 : 154 + 114 + 62 + 252 + 112 + 250 + 170 + 116 =

= 1224

Now

Output :

$$C = \begin{bmatrix} 255 & 258 & 250 & 209 \\ 458 & 566 & 552 & 472 \\ 708 & 981 & 887 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

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